Alexandre Grothendieck

Buffalo Lectures on Topos Theory

1973

Index of Results

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Introduction

The course. From April to July 1973, Alexandre Grothendieck gave a course on topos theory, in English, at the State University of New York at Buffalo, alongside two other courses, on algebraic geometry and on algebraic groups. The sessions were recorded on magnetic tape. This document relies on the recordings published by the YouTube channel "Erwin Schulhoff" in four videos; it cites each session by its date, tape and side, following the descriptions of these videos, and each moment by its position in the video.

What this index is. For each result, definition or example treated in a session, the index gives one line: the statement, in today's mathematical language and in our own words, and the moment on the tape where it is treated, as a link to the video accurate to the second. An entry corresponds to a statement, not to a minute of the lecture. A few words in quotation marks reproduce a name or a term coined in the course, when it matters. When what is said is sketched or incomplete, the entry gives the correct statement, with the hypotheses it requires; a claim that is false as heard is not reproduced. A passage whose audio remains doubtful is not indexed.

What it is not. This index is neither a transcription nor a summary of the course. It reproduces neither the explanations, nor the worked examples, nor the digressions, nor the order of the exposition beyond the succession of statements, and it does not quote the course. The course as delivered is a protected oral work, whose rights belong to the rightsholders of Alexandre Grothendieck; only the recordings are authoritative, and the reader is referred to them through the links. The statements here are standard mathematical statements, written by us.

What is missing. The five sessions of the second video (30 May, 4, 6, 20 and 27 June 1973) are not yet indexed: they are marked *forthcoming*. Four sides whose sound is too degraded are not indexed either (4 April, tape 1, side B; 11 May, tape 2, side A; and two sides without label or date at the end of the fourth video).

1 Presheaves and sheaves on a space

Session of 4 April 1973

Video 1/4 (`URU5jgTvjhU`): tape 1, side A (0:00:00–0:49:11); tape 1, side B (0:49:11–1:36:58); tape 2, side A (1:36:58–2:28:43); tape 2, side B (2:28:43–2:33:29).

Tape 1, side A

`0:03:26` The open sets $\mathcal{O}(X)$ of a space X, ordered by inclusion, form a category: $\mathrm{Hom}(U, V)$ is a singleton if $U \subset V$, empty otherwise.

`0:05:35` Definition: a presheaf of sets on X is a functor $F\colon \mathcal{O}(X)^{\mathrm{op}} \to \mathbf{Set}$; the maps $F(V) \to F(U)$, $U \subset V$, are the restrictions.

`0:07:46` A morphism of presheaves $F \to G$ is a natural transformation: maps $F(U) \to G(U)$ compatible with the restrictions.

`0:09:56` Definition: F is separated if, for every open covering $U = \bigcup U_i$, the map $F(U) \to \prod_i F(U_i)$ is injective.

`0:14:19` Definition: F is a sheaf if, for every covering $U = \bigcup U_i$, the sequence $F(U) \to \prod_i F(U_i) \rightrightarrows \prod_{i,j} F(U_i \cap U_j)$ is exact (equalizer).

0:16:28 The sheaves on X form a full subcategory $\mathrm{Sh}(X)$ (denoted here "Top(X)") of the category of presheaves.

2 Examples of sheaves

0:18:47 Example: for a set E, the presheaf $U \mapsto E^U$ of maps from U to E is a sheaf.

0:20:54 Definition: a subpresheaf $G \subset F$ is a family of subsets $G(U) \subset F(U)$ stable under the restrictions.

0:24:12 A subpresheaf of a separated presheaf is separated.

0:24:12 A subpresheaf G of a sheaf F is a sheaf if and only if membership of a section in G is a local property on U.

0:26:19 Example: if E is a topological space, the continuous maps from open sets of X to E form a subsheaf of $U \mapsto E^U$.

0:27:23 Example: for differentiable (or analytic) manifolds X and E, the differentiable (resp. analytic) maps form a sheaf.

0:29:29 Example: for $p\colon E \to X$ continuous, the continuous sections $\Gamma(U, E)$ over the open sets U form a sheaf.

0:32:45 The sheaf of continuous maps from X to E is isomorphic to the sheaf of sections of the second projection $E \times X \to X$.

0:34:59 Example: on a locally compact space X, the Radon measures on the open sets of X form a sheaf.

0:36:04 Example: on a differentiable manifold, the distributions on the open sets form a sheaf.

0:36:04 Example: $U \mapsto \{\text{subsets of } U\}$, with restriction by intersection, is a sheaf; the closed subsets of U form a subsheaf.

0:38:11 Example: on an analytic space, the analytic subsets of the open sets form a sheaf.

3 Sheaves of sections and the case of a point

0:41:27 The sheaf of sections $\widetilde{E}$ of a space E over X defines a functor $\mathrm{Fib}(X) \to \mathrm{Sh}(X)$, $f \mapsto (s \mapsto f \circ s)$.

0:44:40 This functor is not fully faithful: for X a point, it is identified with the forgetful functor from topological spaces to **Set**.

0:45:45 On a point, the functor $\Gamma\colon F \mapsto F(\mathrm{pt})$ is an equivalence of categories $\mathrm{Sh}(\mathrm{pt}) \simeq$ **Set**.

0:46:51 For U open, $\Gamma_U\colon F \mapsto F(U)$ is a functor $\mathrm{Sh}(X) \to$ **Set**; we write $\Gamma(F) = \Gamma(X, F) = F(X)$.

Tape 1, side B

Not indexed (sound too degraded to be transcribed).

4 Limits and colimits

Tape 2, side A

1:37:06 The notions dual to products, equalizers, fibre products and the terminal object are coproducts, coequalizers, pushouts and the initial object.

1:38:08 In **Set**, coproducts are disjoint unions and the initial object is $\varnothing$.

1:38:08 In **Set**, the coequalizer of $E \rightrightarrows F$ is the quotient of F by the equivalence relation generated by the image of E in $F \times F$.

1:40:15 Example: in groups, the coproduct is the free product; the coproduct of n copies of $\mathbb{Z}$ is the free group on n generators.

1:41:22 For the usual algebraic structures, limits are computed on the underlying sets, but colimits are not.

1:43:32 If S admits limits (resp. colimits) of a given type, $\mathrm{Hom}(C^{\mathrm{op}}, S)$ admits them too, computed argumentwise.

1:44:35 The category opposite to $\mathrm{Hom}(C^{\mathrm{op}}, S)$ is $\mathrm{Hom}(C, S^{\mathrm{op}})$.

1:46:41 Presheaves on X admit all colimits, computed open set by open set.

1:47:49 The subcategory $\mathrm{Sh}(X)$ is not stable in presheaves under any of the usual types of colimit.

1:50:02 Example: the coproduct, taken in presheaves, of the terminal object with itself (the constant presheaf with two elements) is not a sheaf.

1:52:10 For every sheaf F, $F(\varnothing)$ is a singleton ($\varnothing$ being covered by the empty family).

1:53:17 Example: the initial presheaf $U \mapsto \varnothing$ is not a sheaf.

1:54:21 A colimit of sheaves exists if and only if the functor $G \mapsto \mathrm{Hom}(P, G)$ on $\mathrm{Sh}(X)$ is representable, P being the colimit taken in presheaves.

5 The associated sheaf

1:57:33 Definition: the sheaf associated to a presheaf P is a sheaf $\widetilde{P}$ equipped with $P \to \widetilde{P}$, universal among morphisms from P to sheaves.

1:58:36 Theorem: every presheaf admits an associated sheaf, unique up to unique isomorphism.

2:01:50 Corollary: $\mathrm{Sh}(X)$ admits all colimits, the sheaf associated to the colimit computed in presheaves.

2:01:50 A presheaf and its associated sheaf have the same stalks.

2:04:10 In terms of étalé spaces, the coproduct of a family of sheaves is the disjoint topological sum, which is étalé over X.

2:05:16 The coequalizer of $F \rightrightarrows G$ (étalé spaces over X) is G/R, with the quotient topology, R the generated relation; G/R is étalé over X.

2:07:22 Coproducts and coequalizers of double arrows suffice to construct all colimits.

6 Exactness properties of sheaves

2:08:24 Principle: $\mathrm{Sh}(X)$ has the exactness properties of **Set** that involve arbitrary colimits and finite limits.

2:09:28 In **Set**, an arbitrary product of epimorphisms is an epimorphism; this is false in $\mathrm{Sh}(X)$ in general.

2:12:40 Categories of presheaves (of functors to **Set**) have all the exactness properties of **Set**, limits and colimits being computed argumentwise.

2:13:45 The functor $a\colon P \mapsto \widetilde{P}$ is left adjoint to the inclusion i: $\mathrm{Hom}(P, iF) \simeq \mathrm{Hom}(aP, F)$; it commutes with colimits.

2:14:49 Theorem: the associated-sheaf functor is left exact (it commutes with finite products and equalizers).

2:18:08 In $\mathrm{Sh}(X)$, coequalizers are stable under base change: $\mathrm{Coker}(F \rightrightarrows G) \times_H H'$ is the coequalizer of the inverse images over H'.

2:24:32 In $\mathrm{Sh}(X)$, coproducts are stable under base change: $(\coprod_i F_i) \times_F F' \simeq \coprod_i (F_i \times_F F')$.

7 Quotients and equivalence relations

2:27:43 Definition: a quotient object of F is a class of epimorphisms $F \to G$ up to isomorphism under F.

Tape 2, side B

2:28:43 For every morphism $F \to G$, the fibre product $F \times_G F \subset F \times F$ is an equivalence relation on F.

2:29:47 To an equivalence relation $R \subset F \times F$ is associated the coequalizer of $R \rightrightarrows F$, a quotient of F (an epimorphism).

2:30:52 Definition: a quotient is effective if it is the quotient by its relation $F \times_G F$; a relation R is effective if $R = F \times_{F/R} F$.

2:31:57 In $\mathrm{Sh}(X)$, as in **Set**, equivalence relations and quotients are effective and correspond bijectively.

8 Torsors and stalkwise checks

Session of 11 May 1973

Video 1/4 (URU5jgTvjhU): tape 1, side A (2:33:29–3:22:55); tape 1, side B (3:22:55–4:02:08); tape 2, side A (4:02:08–4:48:29); tape 2, side B (4:48:29–5:36:41).

Tape 1, side A

2:33:33 In **Set**, G acts simply transitively on E if $G \times E \to E \times E$, $(g, x) \mapsto (gx, x)$, is bijective; E is a torsor if moreover $E \to \mathrm{pt}$ is surjective.

2:34:34 $E \to F$ is a monomorphism if and only if $E \to E \times_F E$ is an isomorphism; dually, an epimorphism if $F \sqcup_E F \to F$ is an isomorphism.

2:35:38 Definition: a torsor under a sheaf of groups G is a G-sheaf E with $G \times E \simeq E \times E$ and $E \to 1$ an epimorphism (and not $E(U)$ a torsor for every U).

2:36:42 A morphism of sheaves is an epimorphism if and only if it is one on stalks; E is a G-torsor if and only if each E_x is a G_x-torsor.

2:38:50 A torsor is trivial (isomorphic to G acting by translations) if and only if it admits a global section; nontrivial ones exist.

2:40:58 Principle: statements about structures defined by finite limits and colimits can be checked stalkwise, the fibre functors forming an exact conservative family.

2:45:00 The structures of field and of integral domain cannot be expressed by finite limits alone.

2:50:29 A sheaf of rings K is called integral if $K' \sqcup 1 \to K$ (inclusion and zero section) is an epimorphism, K' being defined below.

2:52:38 For two sheaves K, L, the presheaf $\underline{\mathrm{Hom}}(K, L) \colon U \mapsto \mathrm{Hom}(K|_U, L|_U)$ is a sheaf.

2:53:43 The sections f of K whose left and right multiplications are monomorphisms of $K|_U$ form a subsheaf K' (a local property).

9 Generating families and images

2:59:16 Definition: (Z_i) is a generating family of C if every monomorphism $E \to F$ such that $\mathrm{Hom}(Z_i, E) \to \mathrm{Hom}(Z_i, F)$ is bijective for all i is an isomorphism.

3:01:24 In $\mathrm{Sh}(X)$, (Z_i) is generating if and only if the family of functors $\mathrm{Hom}(Z_i, -)$ is conservative.

3:03:48 In $\mathrm{Sh}(X)$, every morphism factors, essentially uniquely, as an epimorphism followed by a monomorphism (image).

3:05:57 In $\mathrm{Sh}(X)$, a morphism that is both a monomorphism and an epimorphism is an isomorphism.

3:09:09 An open set U of X defines a sheaf: $U(V)$ is a singleton if $V \subset U$, empty otherwise.

3:11:24 For every sheaf E, $\mathrm{Hom}(U, E) = E(U)$; the open sets of X therefore form a small generating family of $\mathrm{Sh}(X)$.

10 Abelian sheaves and injectives

3:12:35 The abelian sheaves on X (abelian group objects of $\mathrm{Sh}(X)$) form an abelian category $\mathrm{Ab}(X)$.

3:14:47 $\mathrm{Ab}(X)$ admits limits and colimits; filtered colimits there commute with the forgetful functor and are exact.

3:16:56 Definition: an abelian sheaf I is injective if every morphism $F \to I$ extends along every monomorphism $F \to G$.

3:18:01 An abelian category with exact filtered colimits and a small generating family has enough injectives; this is the case of $\mathrm{Ab}(X)$ (free abelian sheaves on the Z_i).

3:19:05 For a sheaf of rings A, $\mathrm{Mod}(A)$ is abelian, with exact filtered colimits and generators, hence has enough injectives (Ext, cohomology).

3:21:12 In general $\mathrm{Ab}(X)$ does not have enough projectives; filtered limits there are not exact.

11 Direct and inverse images

Tape 1, side B

3:24:08 Definition: for $f\colon X \to Y$, an f-morphism $G \to F$ is a family $G(V) \to F(U)$, for $f(U) \subset V$, compatible with the restrictions.

3:27:20 An f-morphism is equivalent to a family $G(V) \to F(f^{-1}(V))$ compatible with the restrictions.

3:30:33 $V \mapsto f^{-1}(V)$ is an increasing functor $\mathcal{O}_Y \to \mathcal{O}_X$ that commutes with arbitrary unions.

3:31:42 Definition: direct image of a presheaf F on X, $f_*F(V) = F(f^{-1}(V))$, a presheaf on Y.

3:32:48 The f-morphisms $G \to F$ correspond bijectively to the morphisms $G \to f_*F$ of presheaves on Y.

3:33:52 On presheaves, f_* commutes with arbitrary limits and colimits.

3:34:59 The direct image of a sheaf is a sheaf.

3:38:17 $f_*\colon \mathrm{Sh}(X) \to \mathrm{Sh}(Y)$ commutes with arbitrary limits.

3:39:22 In general f_* commutes neither with coproducts nor with epimorphisms: for Y a point, f_* is $\Gamma(X, -)$.

3:41:34 Definition: the inverse image f^*G is equipped with a universal f-morphism $G \to f^*G$; f^* is the left adjoint of f_*.

3:44:45 For every functor $\varphi\colon C \to D$, the restriction $\widehat{D} \to \widehat{C}$, $P \mapsto P \circ \varphi$, has a left adjoint and a right adjoint.

3:46:54 f^*G is the sheaf associated to the presheaf inverse image $f^*_{\mathrm{pr}}G$.

3:49:03 f^* commutes with arbitrary colimits (left adjoint).

3:50:13 Theorem: f^* is left exact (it commutes with finite limits).

3:52:25 The inverse image of a space $G \to Y$ under f is $X \times_Y G \subset X \times G$, with the induced topology; it is a fibre product in topological spaces.

3:53:31 The property of being étalé is stable under base change.

3:54:35 For sheaves viewed as étalé spaces, f^*G corresponds to the fibre product $X \times_Y G$.

3:57:48 Étalé spaces over Y are stable under finite limits in spaces over Y.

3:57:48 The fibre of $X \times_Y G$ at x is homeomorphic to the fibre of G at $f(x)$.

3:58:52 $(f^*G)_x \simeq G_{f(x)}$; hence again the left exactness of f^*.

Tape 2, side A

Not indexed (sound too degraded to be transcribed).

12 Stalks and epimorphisms

Tape 2, side B

4:48:31 Two morphisms $u, v\colon F \to G$ of sheaves are equal if and only if $u_x = v_x$ for all x: the fibre functors form a faithful family.

4:50:40 The fibre functors commute with colimits and finite limits and form a conservative family.

4:53:05 $F \to G$ is an epimorphism of sheaves if and only if every section of G lifts locally to a section of F.

4:55:12 An epimorphism of sheaves is not in general an epimorphism of presheaves: $\Gamma(F) \to \Gamma(G)$ is not always surjective.

4:55:12 For X compact, $\Gamma(X, -)$ sends epimorphisms of sheaves to surjections if and only if X is totally disconnected.

13 Algebraic structures on objects of a category

4:59:30 Definition: a composition law on an object E of C is an arrow $E \times E \to E$; commutativity and associativity are expressed by diagrams.

5:01:42 A left unit is an arrow $1 \to E$ (1 terminal) such that $E \simeq 1 \times E \to E \times E \to E$ is the identity.

5:04:55 Via the Yoneda embedding, a law $E \times E \to E$ is equivalent to laws on the $E(Z) = \mathrm{Hom}(Z, E)$, functorial in Z.

5:07:07 Definition: an algebraic structure (group, ring, module, etc.) on E is such a structure on each $E(Z)$, functorial in Z.

5:11:27 The structure of a category ($D_1 \rightrightarrows D_0$, composition $D_1 \times_{D_0} D_1 \to D_1$, units) is described by fibre products.

5:13:34 Definition: a category internal to C is a pair (D_0, D_1) with a category structure on $(D_0(Z), D_1(Z))$, functorial in Z.

5:14:41 The embedding $C \to \widehat{C}$ commutes with arbitrary limits, but not with colimits in general.

5:15:45 A structure defined by limits on a sheaf F is equivalent to the same structure on each $F(U)$, functorial in U.

5:17:53 A sheaf of groups is a functor $\mathcal{O}(X)^{\mathrm{op}} \to \mathbf{Grp}$ whose composite with the forgetful functor is a sheaf.

5:22:21 The axioms of a structure defined by finite limits can be checked stalkwise.

5:23:26 A functor that commutes with finite products sends group objects (ring objects, etc.) to objects of the same type.

5:25:32 The stalks of a sheaf of groups (of rings, etc.) are groups (rings, etc.).

5:27:45 Definition: for a k-coalgebra A, an A-comodule is a k-module M equipped with $M \to M \otimes_k A$ satisfying coassociativity and counit.

5:30:56 The tensor product of modules is constructed by colimits (sums, quotients), not by limits alone.

5:33:19 For sheaves of modules, $M(U) \otimes_{k(U)} A(U) \to (M \otimes_k A)(U)$ is in general neither injective nor surjective.

5:34:24 Definition: E is a torsor under a group G if $E \neq \varnothing$ and G acts simply transitively on E.

14 Étale morphisms and slice categories

Session of 18 May 1973

Video 1/4 (URU5jgTvjhU): tape 1, side A (5:36:41–6:26:00); tape 1, side B (6:26:00–7:14:32); tape 2, side A (7:14:32–8:03:53); tape 2, side B (8:03:53–8:23:34).

Tape 1, side A

5:39:20 For a point y of Y and the inclusion $i_y : \{y\} \to Y$, we have $i_y^* F \simeq F_y$: the stalk is the inverse image under a point.

5:41:31 If $f : X \to Y$ is étale, X is an object of $\mathrm{Sh}(Y)$ and $\mathrm{Sh}(X) \simeq \mathrm{Sh}(Y)/X$; a composite of étale morphisms is étale.

5:43:40 Definition: for an object X of a category C, the category C/X has as objects the arrows $X' \to X$ and as morphisms the commutative triangles.

5:44:44 If $f : X \to Y$ is étale, f^*G is the product $G \times X$ in $\mathrm{Sh}(Y)$, viewed as an object of $\mathrm{Sh}(Y)/X$.

5:47:00 For $g : X' \to X$ in $\mathrm{Sh}(Y)$, the functor $g^* : \mathrm{Sh}(Y)/X \to \mathrm{Sh}(Y)/X'$ is the base change $F \mapsto F \times_X X'$.

5:51:15 f_* sends every structure defined by limits (group, ring, etc.) to a structure of the same kind, but not torsors.

5:52:19 f^* respects structures defined by finite limits and colimits: the inverse image of a G-torsor is an f^*G-torsor.

15 Ring and field objects

5:56:33 Definition: a ring object A of C is an object equipped with a ring structure on each $A(Z) = \mathrm{Hom}_C(Z, A)$, functorial in Z.

5:57:38 The units $A^*(Z) \subset A(Z)$ form a group-valued functor, represented by the fibre product of $m : A \times A \to A$ and $1 : e \to A$.

5:59:47 Definition: a ring object A is a field object if the morphism $A^* \sqcup e \to A$ (inclusion and zero section) is an isomorphism.

6:02:58 Integrality, first condition: for $C = m^{-1}(0) \subset A \times A$, the morphism $A \sqcup A \to C$, $x \mapsto (x, 0)$, $x \mapsto (0, x)$, is an epimorphism.

6:07:15 A morphism $S \to C$ is an epimorphism iff the canonical arrow $C \sqcup_S C \to C$ is an isomorphism.

6:08:19 Integrality, second condition ($0 \neq 1$): the kernel $e' \subset e$ of $0, 1 : e \rightrightarrows A$ is initial; in $\mathrm{Sh}(X)$, e' is the largest open set where $A = 0$.

6:11:32 Every kind of structure defined by finite limits and arbitrary colimits admits a "classifying topos", equipped with a universal model.

6:12:37 Normal rings, reduced rings, rings with connected spectrum can also be defined by such diagrammatic axioms.

6:13:40 A sheaf of rings is never a presheaf of fields: its value on $\varnothing$ is the zero ring.

6:15:45 On a space, such an axiom is checked stalkwise: a sheaf of rings A is a field object iff each stalk A_x is a field.

16 Constant sheaves and sheaves of fields

6:16:49 The constant sheaf $I_X = p^*I$ ($p : X \to \mathrm{pt}$) has étalé space $I \times X$; it is the coproduct of I copies of the terminal object.

6:19:00 If I is a group, a ring or a field, I_X is a sheaf of groups, of rings or of fields; for example K_X for a field K.

6:20:05 On a reduced and locally irreducible complex analytic space, the sheaf $\mathcal{M}_X$ of meromorphic functions is a sheaf of fields.

6:21:07 The stalk $\mathcal{M}_{X,x}$ is the total ring of fractions of $\mathcal{O}_{X,x}$; it is a field as soon as $\mathcal{O}_{X,x}$ is a domain.

6:22:14 On an integral scheme, the sheaf of rational functions is a constant sheaf of fields; the analytic sheaf $\mathcal{M}_X$ is not.

17 Internal Hom and the subobject sheaf

6:23:18 Definition: $\mathcal{H}om(F,G)(U) = \mathrm{Hom}(F|_U, G|_U)$ defines a sheaf on X, to be distinguished from the set $\mathrm{Hom}(F,G)$.

Tape 1, side B

6:26:09 For X' étale over X: $\mathrm{Hom}(X', \mathcal{H}om(F,G)) \simeq \mathrm{Hom}_{X'}(F \times X', G \times X') \simeq \mathrm{Hom}(F \times X', G)$.

6:29:26 Definition: in a category with products, $\mathcal{H}om(F,G)$ is, if it exists, the object representing $X' \mapsto \mathrm{Hom}(F \times X', G)$; it exists in $\mathrm{Sh}(X)$.

6:30:28 For group objects F, G, $\mathcal{H}om_{\mathrm{gr}}(F,G)$ represents $X' \mapsto \mathrm{Hom}_{\mathrm{gr}}(F_{X'}, G_{X'})$, homomorphisms of group objects of C/X'.

6:32:35 The sheaf $\mathcal{P}(F)$: $\mathcal{P}(F)(U)$ is the set of subsheaves of $F|_U$; for X a point, it is the set of subsets of F.

6:34:44 $\mathcal{P}(F)$ represents $X' \mapsto \mathrm{Sub}(F \times X')$; the subobjects of an object of C/X' are the same in C and in C/X'.

6:38:01 For the terminal object e, $\mathcal{P}(e)$ is the sheaf that associates to U the set of its open subsets; it represents $X' \mapsto \mathrm{Sub}(X')$.

6:41:10 If $\mathcal{P}(e)$ exists, $\mathcal{P}(F)$ exists iff $\mathcal{H}om(F, \mathcal{P}(e))$ exists, and they are isomorphic; in **Set**, $\mathcal{P}(e)$ has two elements.

6:44:25 A subsheaf of $\mathcal{P}(F)$ is a family of sets of subsheaves of the $F|_U$, stable under restriction and local in U.

6:46:35 A topology on a sheaf F is defined as a subsheaf of $\mathcal{P}(F)$ satisfying the usual axioms, read in $\mathrm{Sh}(X)$.

6:47:39 For $f : X \to Y$, there are canonical arrows $f^*\mathcal{H}om(F,G) \to \mathcal{H}om(f^*F, f^*G)$ and $f^*\mathcal{P}(F) \to \mathcal{P}(f^*F)$.

6:50:49 These arrows are not isomorphisms in general, for example for a fibre functor; they are if f is étale.

18 Stalks and infinite products

6:51:54 f^* does not commute with infinite products in general; there is a canonical arrow $(\prod_i F_i)_x \to \prod_i (F_i)_x$.

6:54:11 Its injectivity means: if f_i, g_i have the same germ at x for every i, they coincide on a common neighbourhood of x.

6:55:19 Counterexample: U_i open neighbourhoods of x whose intersection is not a neighbourhood, F the sheaf of maps to $\{a, b\}$, $f_i = a$, $g_i = a$ on U_i and b elsewhere.

6:57:25 The stalk functor at x commutes with products iff x has a smallest open neighbourhood; if points are closed, iff x is isolated.

6:58:33 In $\mathrm{Sh}(X)$, an infinite product of epimorphisms is not always an epimorphism; one reduces to the case of the $F_i \to e$.

7:01:47 Example: X locally connected, x closed, U_i open neighbourhoods of x whose intersection is not a neighbourhood, $F_i = U_i \sqcup (X \setminus \{x\})$; $\prod_i F_i \to e$ is not an epimorphism.

7:05:03 If X is compact and totally disconnected, $F \to G$ is an epimorphism iff $F(U) \to G(U)$ is surjective for every clopen set U; every product of epimorphisms is an epimorphism.

19 Connectedness and cohomology

7:12:57 X is connected iff the terminal object of $\mathrm{Sh}(X)$ is not a sum of two non-initial subobjects.

Tape 2, side A

7:14:35 X is nonempty iff $\varnothing \to e$ is not an isomorphism in $\mathrm{Sh}(X)$; X is empty iff $\mathrm{Sh}(X)$ is equivalent to the point category.

7:16:42 The abelian group objects of $\mathrm{Sh}(X)$ form a category with enough injectives; $H^i(X,F) = R^i\Gamma(F)$, with $\Gamma(F) = \mathrm{Hom}(e, F)$.

7:18:48 For an abelian group M, we set $H^i(X,M) = H^i(X, M_X)$; these groups depend only on the category $\mathrm{Sh}(X)$.

7:19:52 X is connected iff $\mathbb{Z} \to H^0(X,\mathbb{Z})$ is surjective; connected and nonempty iff it is bijective.

7:20:57 The sections of I_X over U are the locally constant maps $U \to I$.

7:23:07 The lattice of open sets of X is that of the subobjects of the terminal object of $\mathrm{Sh}(X)$; the Čech constructions (Artin–Mazur) follow from it.

20 Covering spaces and torsors

7:24:10 If X is connected, locally connected and locally simply connected, the covering spaces of X correspond to $\pi_1(X,x)$-sets.

7:25:16 The covering spaces of X are the locally constant objects X' of $\mathrm{Sh}(X)$: $X' \times U_i \simeq \coprod_{S_i} U_i$ for a covering $e = \sup_i U_i$.

7:26:21 For X connected and locally connected but not locally simply connected, the covering spaces are classified by a pro-group.

7:27:24 For a sheaf of groups G not necessarily commutative, $H^1(X,G)$ is the set of isomorphism classes of G-torsors.

7:28:31 Definition: a G-torsor is an object P equipped with an action of G such that $G\times P \to P \times P$ is an isomorphism and $P \to e$ an epimorphism.

7:29:35 $G \mapsto G_X = \coprod_G e$ is left exact; the G_X-torsors are the Galois covering spaces with group G.

7:30:40 For X connected, locally connected and locally simply connected, $H^1(X,G)$ is identified with $\mathrm{Hom}(\pi_1(X,x),G)$ modulo conjugation; X is simply connected iff all these H^1 are trivial.

21 Stacks and gerbes

7:31:44 Definition: a fibred category over $\mathcal{O}(X)$ associates to U a category F_U, and to $V \subset U$ a restriction functor, transitive up to coherent isomorphism.

7:32:47 Examples (which are stacks): $U \mapsto \mathrm{Sh}(U)$, fibre spaces over U, sheaves of groups on U, $G|_U$-torsors.

7:34:51 Definition: a fibred category is a stack if the presheaves $\mathcal{H}om(F,G)$ are sheaves and if every gluing datum (cocycle on the U_{ijk}) is effective.

7:39:09 For G noncommutative, $H^2(X,G)$ is defined as a set of classes of stacks (gerbes) attached to G (Giraud); it depends only on $\mathrm{Sh}(X)$.

7:44:33 Definition: a stack over $\mathcal{E}$ is a fibred category over $\mathcal{E}$ with descent along epimorphic families, the intersections becoming fibre products.

7:45:38 Restriction and extension give an equivalence between stacks over $\mathcal{O}(X)$ and stacks over $\mathrm{Sh}(X)$.

22 Gluing and quotient topoi

7:48:59 $\mathrm{Sh}(\coprod_i X_i) \simeq \prod_i \mathrm{Sh}(X_i)$: the sum of spaces corresponds to the product of categories.

7:50:03 If Z is closed in X and in Y, $\mathrm{Sh}(X \cup_Z Y)$ is the 2-fibre product of $\mathrm{Sh}(X)$ and $\mathrm{Sh}(Y)$ over $\mathrm{Sh}(Z)$.

7:52:11 For an equivalence relation R on X, the sheaves on X equipped with descent data relative to R form a quotient topos, even if X/R is pathological.

23 Irreducible closed sets and sober spaces

7:54:19 Definition: a space Z is irreducible if it is nonempty and not a union of two distinct closed subsets of Z.

7:55:24 A separated space is irreducible iff it is a point; a variety is irreducible for the Zariski topology iff it is so in the algebraic sense.

7:55:24 The closure of an irreducible subset is irreducible.

7:56:30 We write $\tilde{X}$ for the set of irreducible closed subsets of X and $i : X \to \tilde{X}$, $x \mapsto \overline{\{x\}}$; i is bijective if X is separated.

7:57:34 The subsets $\tilde{U} = \{Y \in \tilde{X} : Y \cap U \neq \varnothing\}$, U open, form a topology: $U \mapsto \tilde{U}$ respects unions and finite intersections.

8:00:50 We have $i^{-1}(\tilde{U}) = U$; i is continuous and induces a lattice isomorphism $\mathcal{O}(\tilde{X}) \simeq \mathcal{O}(X)$.

8:03:02 Consequently $i_* : \mathrm{Sh}(X) \to \mathrm{Sh}(\tilde{X})$ is an isomorphism of categories, with inverse i^*.

Tape 2, side B

8:03:56 $i : X \to \tilde{X}$ is injective iff X is T_0; if X is separated, i is a homeomorphism.

8:05:01 Definition: a generic point of a closed set Z is a point whose closure is Z; i is surjective iff every irreducible closed set has a generic point.

8:06:06 Example: a variety over K reduced to its K-rational points has irreducible closed sets without a generic point.

8:07:14 Definition: X is sober if every irreducible closed set has exactly one generic point, i.e. if $i : X \to \tilde{X}$ is bijective, hence a homeomorphism.

8:08:20 For Y an irreducible closed set, the stalk $F_Y = \varinjlim_{U \cap Y \neq \varnothing} F(U)$ (filtered colimit) is left exact and commutes with colimits.

8:10:27 $F_{\overline{\{x\}}} = F_x$; in general F_Y is the stalk at the point Y of F viewed as a sheaf on $\tilde{X}$.

8:11:30 $Y \mapsto (F \mapsto F_Y)$ is a bijection from $\tilde{X}$ onto the classes of fibre functors $\mathrm{Sh}(X) \to \mathbf{Set}$ (left exact, commuting with $\varinjlim$).

8:12:39 A fibre functor on $\mathrm{Sh}(X)$ has no nontrivial automorphism.

8:13:42 $\tilde{X}$ is sober, and $X \mapsto \tilde{X}$ is left adjoint to the inclusion of sober spaces: $\mathrm{Hom}(X, Y) \simeq \mathrm{Hom}(\tilde{X}, Y)$ for Y sober.

8:16:56 On the set $\Phi(\mathcal{E})$ of classes of fibre functors, the sets $\tilde{U} = \{\varepsilon : \varepsilon(U) \neq \varnothing\}$, $U \subset e$, form a topology.

8:20:13 $X \to \Phi(\mathrm{Sh}(X))$ is a homeomorphism iff X is sober: a sober space can be reconstructed from $\mathrm{Sh}(X)$.

8:21:16 If X is T_1 (for example the closed points of a Jacobson scheme), X is identified with the subspace of closed points of $\tilde{X}$.

24 Points and morphisms of sober spaces

Session of 25 May 1973

Video 1/4 (URU5jgTvjhU): tape 1, side A (8:23:34–9:28:43); tape 1, side B (9:28:43–10:33:42); tape 2, side A (10:33:42–11:06:44).

Tape 1, side A

8:24:40 For $f : X \to Y$ continuous, the point $f(x)$ corresponds to the fibre functor $\varepsilon_x \circ f^*$, since $(f^*F)_x \simeq F_{f(x)}$.

8:27:51 If X and Y are sober, every functor $\mathrm{Sh}(Y) \to \mathrm{Sh}(X)$ that is left exact and commutes with $\varinjlim$ is isomorphic to f^* for a unique continuous f.

8:29:59 Two functors on $\mathrm{Sh}(Y)$ that commute with colimits and are isomorphic on the generating subcategory $\mathcal{O}(Y)$ are isomorphic.

8:33:13 For $f, g : X \to Y$, a morphism from f to g is a morphism between f^* and g^*; if Y is separated, there is one only if $f = g$, and it is the identity.

25 Definitions of a topos

8:37:46 Definition (first form): a topos is a category having the exactness properties of **Set** (arbitrary $\varinjlim$, finite $\varprojlim$) and a small generating set.

8:38:52 Definition: $C \subset \mathrm{Ob}\,\mathcal{E}$ is generating if $\mathcal{E} \to \widehat{C}$, $X \mapsto \mathrm{Hom}(-, X)|_C$, is faithful.

8:40:58 In a topos, it amounts to the same whether this functor is faithful, fully faithful, conservative, or conservative on monomorphisms.

8:44:06 Definition (Giraud): a topos is a category with (a) finite $\varprojlim$, (b) universal disjoint coproducts, (c) universal effective equivalence relations, (d) a small generator.

8:46:13 Definition: a coproduct $X = \coprod X_i$ is universal if for every $X' \to X$, the arrow $\coprod_i (X_i \times_X X') \to X'$ is an isomorphism.

8:48:24 Definition: it is disjoint if the $X_i \to X$ are monomorphisms and $X_i \times_X X_j$ is initial for $i \neq j$.

8:51:36 Definition: an equivalence relation $R \subset X \times X$ (each $R(Z)$ being one on $X(Z)$) is effective if $R \simeq X \times_{X/R} X$.

8:52:42 It is universal if $X \to X/R$ is a universal epimorphism and if passage to the quotient commutes with base changes $Y' \to X/R$.

8:55:52 In a topos, the subobjects and the quotient objects of an object form small sets.

26 Cohomology in a topos

8:56:57 $\mathcal{E}_{ab}$ is abelian, with exact filtered colimits and a generator; it therefore has enough injectives.

8:58:03 An injective object of $\mathcal{E}_{ab}$ is injective in $\mathcal{E}$; in particular $\mathcal{E}$ has enough injectives.

9:02:21 In a topos, $\Gamma(X) = \mathrm{Hom}(e, X)$; its derived functors on $\mathcal{E}_{ab}$ are the $H^i(\mathcal{E}, -)$.

9:04:28 The constant object $I_{\mathcal{E}} = \coprod_I e$: $I \mapsto I_{\mathcal{E}}$ commutes with $\varinjlim$ and finite $\varprojlim$; we set $H^i(\mathcal{E}, M) = H^i(\mathcal{E}, M_{\mathcal{E}})$.

9:06:45 A topos has homotopy invariants (pro-groups, the Čech method of Artin–Mazur), which recover the π_i of a locally contractible space.

27 Basic examples of topoi

9:09:57 $\mathrm{Sh}(\varnothing)$ is equivalent to the point category: it is the "empty topos".

9:12:01 A topos $\mathcal{E}$ is empty iff the arrow $\varnothing_{\mathcal{E}} \to e_{\mathcal{E}}$ is an isomorphism.

9:14:08 $\mathrm{Sh}(\mathrm{pt}) \simeq \mathbf{Set}$ via $F \mapsto F(\mathrm{pt})$: it is the point topos, or "final topos".

9:17:23 A topos $\mathcal{E}$ is a point iff the functor $\mathbf{Set} \to \mathcal{E}$, $I \mapsto I_{\mathcal{E}}$, is an equivalence.

9:18:25 For a discrete group G acting on X, the category $\mathrm{Sh}(X, G)$ of étalé spaces over X equipped with a compatible G-action is a topos.

9:21:38 $\mathrm{Sh}(X, \{1\}) = \mathrm{Sh}(X)$; $\mathrm{Sh}(\mathrm{pt}, G) = B_G$, the category of G-sets, is the "classifying topos" of G.

9:23:42 $(B_G)_{ab}$ is the category of G-modules, $\Gamma(M) = M^G$ and $H^i(B_G, M) = H^i(G, M)$, the cohomology of the classifying space of G.

9:25:54 The cohomology of $\mathrm{Sh}(X, G)$ is the equivariant cohomology $H^i(X; G, M)$ (Tôhoku).

28 Points of a topos

Tape 1, side B

9:28:50 If $G \neq 1$, the topos B_G is equivalent to no $\mathrm{Sh}(X)$.

9:30:01 The forgetful functor $B_G \to \mathbf{Set}$ is a fibre functor, with automorphism group G.

9:32:08 If X is sober, every fibre functor on $\mathrm{Sh}(X)$ is isomorphic to $F \mapsto F_x$ for a unique $x \in X$.

9:33:13 Specialization ($y \in \overline{\{x\}}$) is a preorder, which is an order if X is sober; it induces $F_y \to F_x$, functorial in F.

9:37:42 There is a morphism $\varepsilon_y \to \varepsilon_x$ iff y is a specialization of x, and it is unique: the points of $\mathrm{Sh}(X)$ form an ordered set.

9:38:48 B_G has a single point up to isomorphism without being the final topos: its points have nontrivial automorphisms.

9:40:58 In $\mathrm{Sh}(X, G)$, $\varepsilon_x^G : F \mapsto F_x$ is a point, with automorphism group the stabilizer G_x.

29 Giraud's theorem and elementary topoi

9:46:33 Theorem (Giraud): $\mathcal{E}$ is a topos iff there exist C small and $i : \mathcal{E} \to \widehat{C}$ fully faithful with a left exact left adjoint.

9:51:03 For every generating full subcategory C of a topos, $\mathcal{E} \to \widehat{C}$ is fully faithful and has a left exact left adjoint.

9:53:11 Topoi in the sense of Giraud's axioms are exactly the categories of sheaves on a site.

9:55:24 In a topos, $\mathcal{H}om(X, Y)$ exists: it represents $Z \mapsto \mathrm{Hom}(X \times Z, Y)$.

9:56:29 In a topos, the functor $Z \mapsto \mathrm{Sub}(Z)$ is representable by an object $P_{\mathcal{E}}$.

9:57:40 Lemma: in a topos, $P : \mathcal{E}^\circ \to \mathbf{Set}$ is representable iff it sends colimits to limits.

10:00:55 Every topos is an elementary topos (Lawvere–Tierney: finite $\varprojlim$, $\mathcal{H}om$, $P_{\mathcal{E}}$); an elementary topos with small coproducts and a small generator is a topos.

30 Presheaf topoi and the classifying topos of a group

10:05:07 In $\mathrm{Sh}(X, G)$, limits, $\mathcal{H}om$ and P are computed in $\mathrm{Sh}(X)$, with the induced action of G.

10:07:16 For C small, $\widehat{C}$ is a topos (limits argumentwise, representables generating), with all the exactness properties of **Set**.

10:10:30 $B_G = \widehat{C}$ for C the category with one object whose endomorphisms are G.

10:11:33 For $X \in C$, $F \mapsto F(X)$ is a point of $\widehat{C}$; every point is a filtered colimit of these, and $\widehat{C}$ has enough points.

10:12:36 $\mathrm{Sh}(X, G)$ has enough points: the ε_x^G form a conservative family.

10:14:42 For a group object G of a topos $\mathcal{E}$, the objects of $\mathcal{E}$ equipped with an action of G form a topos B_G; for $\mathcal{E} = \mathbf{Set}$, the G-sets.

10:17:55 $H^i(G, M) = H^i(B_G, M)$ for M abelian with an action of G; $\Gamma(X)$ is the set of G-invariant sections of X.

10:21:11 B_G classifies G-torsors: for $f : \mathcal{T} \to \mathcal{E}$, the morphisms $\mathcal{T} \to B_G$ over $\mathcal{E}$ correspond to the f^*G-torsors.

31 Induced topoi

10:23:20 For an object X of a topos $\mathcal{E}$, $\mathcal{E}/X$ is a topos, the induced topos; $\mathrm{Sh}(Y)/X \simeq \mathrm{Sh}(X)$ for X étale over Y.

32 Morphisms of topoi

10:32:10 Between topoi, every functor f^* commuting with colimits has a right adjoint f_* (representability lemma).

Tape 2, side A

10:35:54 Definition: a morphism of topoi $f : \mathcal{E} \to \mathcal{F}$ is a pair of adjoint functors $f^* \dashv f_*$, with $f^* : \mathcal{F} \to \mathcal{E}$ left exact.

10:38:05 The morphisms $\mathcal{E} \to \mathcal{F}$ form a category $\mathbf{Homtop}(\mathcal{E}, \mathcal{F})$, whose arrows $f \to g$ are the morphisms $f_* \to g_*$ (or $g^* \to f^*$).

10:45:28 Composition: $(gf)_* = g_* f_*$ and $(gf)^* = f^* g^*$; it is associative.

10:47:39 Composition is a functor $\mathbf{Homtop}(\mathcal{E}, \mathcal{F}) \times \mathbf{Homtop}(\mathcal{F}, \mathcal{G}) \to \mathbf{Homtop}(\mathcal{E}, \mathcal{G})$: topoi form a 2-category.

10:53:57 A morphism f is an equivalence of topoi iff f^*, or f_*, is an equivalence of categories.

10:59:20 For Y sober, the category of points $\mathbf{Homtop}(\mathrm{pt}, \mathrm{Sh}(Y))$ is equivalent to Y ordered by specialization.

11:01:29 For X, Y sober, $\mathbf{Homtop}(\mathrm{Sh}(X), \mathrm{Sh}(Y))$ is equivalent to the continuous maps f ordered by $f \leq g$ iff $g(x) \in \overline{\{f(x)\}}$ for all x.

11:04:45 If Y is separated, this category is discrete.

33 Sessions not yet indexed

Session of 30 May 1973

Video 2/4 (`PqCQM3Pcl6Q`): tape 1, side A (0:00:00–0:49:07); tape 1, side B (0:49:07–1:38:30); tape 2, side A (1:38:30–1:55:37).

Forthcoming: this session has not yet been transcribed.

Session of 4 June 1973

Video 2/4 (`PqCQM3Pcl6Q`): tape 1, side A (1:55:37–2:44:38); tape 1, side B (2:44:38–3:33:35); tape 2 (3:33:35–4:22:54).

Forthcoming: this session has not yet been transcribed.

Session of 6 June 1973

Video 2/4 (`PqCQM3Pcl6Q`): tape 1 (4:22:54–5:12:09).

Forthcoming: this session has not yet been transcribed.

Session of 20 June 1973

Video 2/4 (`PqCQM3Pcl6Q`): tape 1, side A (5:12:09–6:01:04); tape 1, side B (6:01:04–6:50:09); tape 2, side A (6:50:09–7:38:43); tape 2, side B (7:38:43–8:27:21).

Forthcoming: this session has not yet been transcribed.

Session of 27 June 1973

Video 2/4 (`PqCQM3Pcl6Q`): tape 1, side A (8:27:21–9:02:23); tape 2, side A (9:02:23–9:52:24); tape 2, side B (9:52:24–10:14:40).

Forthcoming: this session has not yet been transcribed.

34 Algebraic theories and strict cogeneration

Session of 2 July 1973

Video 3/4 (`4gynb9uTEE4`): tape 1 (0:00:00–0:48:52); tape 2 (0:48:52–1:54:02); tape 3, side A (1:54:02–2:43:03); tape 3, side B (2:43:03–3:31:57).

Tape 1

`0:04:18` For Δ = categories with arbitrary limits, the localization $C \to C[\Sigma^{-1}]$ taken in Δ is bijective on objects: adding axioms adds arrows, not constructions.

`0:05:21` A category I is strictly cogenerating in its envelope $\widetilde{I}$ under arbitrary limits.

`0:06:26` Definition: $J \subset E$ is "strictly generating" if every object of E is a colimit of a diagram of objects of J; example: J in $\widehat{J}$.

`0:07:31` If I is strictly cogenerating in each R_α of a filtered system in Δ, it is so in its 2-colimit taken in Δ.

`0:10:48` Strict cogeneration is transitive; it is preserved by adjoining arrows in the sense of Δ and by localization in Δ.

0:11:52 Corollary: at each step of the construction of an algebraic theory with base I, the image of I remains strictly cogenerating.

0:12:56 Theorem: for Δ = categories with arbitrary limits, the algebraic theories are those represented by an $R \in \Delta$ having a small strictly cogenerating subcategory.

0:14:08 Moreover, $I \to R$ is basic if and only if the image of I is strictly cogenerating; dually, R^{op} has arbitrary colimits and a small strictly generating set.

0:18:24 If $I' \subset R$ is a small full strictly cogenerating subcategory, the Δ-functor $\widetilde{I'} \to R$ is essentially surjective and full; R is a category of fractions of $\widetilde{I'}$.

0:22:37 Corollary: such a theory is obtained in three steps (empty structure on I, adjoining arrows, axioms), without transfinite construction.

0:30:45 If T is represented by R having a small strictly cogenerating subcategory, $T(\mathbf{Set}) = \mathrm{Hom}_{\Delta}(R, \mathbf{Set}) \simeq R^{\mathrm{op}}$: T is determined by its realizations in sets.

0:37:13 Example: for the theory of groups, $R \simeq \mathbf{Grp}^{\mathrm{op}}$; the universal object is $\mathbb{Z}$, a cogroup of **Grp** via $\mathbb{Z} \to \mathbb{Z} * \mathbb{Z}$, $x \mapsto x_1 x_2$.

0:44:41 For C small and a class D' of diagrams, the envelope of C under limits of type D' is the smallest full subcategory of the envelope $\widetilde{C}$ containing C and stable under these limits.

0:45:46 Example: on categories with finite products, the theory of groups is represented by a small category, the full subcategory of $\mathbf{Grp}^{\mathrm{op}}$ of finitely generated free groups.

0:46:53 A theory on categories with limits of type D' defines by restriction one on categories with arbitrary limits; there is no inverse operation in general.

35 Accessible functors

Tape 2

0:51:15 If $F : E \to E'$ is π-accessible and sends E_π into E'_π (π-accessible objects), F is the extension Ind_π of its restriction $E_\pi \to E'_\pi$.

0:52:15 For F accessible and π large enough, $F(E_\pi) \subset E'_{\pi^{\pi_0}}$ (π_0 fixed, SGA 4); since $(2^\lambda)^{\pi_0} = 2^\lambda$ for $\lambda \geq \pi_0$ infinite, $F(E_\pi) \subset E'_\pi$ if $\pi = 2^\lambda$.

0:54:23 Corollary: an accessible functor between accessible categories respects the canonical filtration by π-accessible objects for arbitrarily large π.

0:56:38 For $i : C \hookrightarrow E$ small full and E' with colimits, the restriction $i^* : \mathrm{Hom}(E, E') \to \mathrm{Hom}(C, E')$ has a fully faithful left adjoint $i_!$.

0:57:40 For E accessible, $F : E \to E'$ is accessible if and only if it is in the essential image of $i_!$ for some small $C \subset E$.

0:59:46 For C small with split idempotents, $\mathrm{Ind}_\pi(C)$ has arbitrary colimits if and only if C has colimits of cardinality at most π.

1:03:11 For such a C, $\mathrm{Ind}_\pi(C)$ is the Δ_π–Δ envelope of C: $\mathrm{Hom}_\Delta(\mathrm{Ind}_\pi C, F) \simeq \mathrm{Hom}_{\Delta_\pi}(C, F)$ for every F with arbitrary colimits.

36 Representability and accessible categories

1:06:26 The categories R representing algebraic theories are exactly the "co-accessible" categories: $S = T(\mathbf{Set}) \simeq R^{\mathrm{op}}$ is accessible (and cocomplete).

1:09:46 On a cocomplete accessible category E, a functor $E^{\mathrm{op}} \to \mathbf{Set}$ is representable if and only if it sends colimits to limits.

1:10:49 A functor $F : E \to \mathbf{Set}$ (E cocomplete accessible) is representable if and only if it commutes with arbitrary limits and is accessible.

1:12:06 SGA 4: a functor $F : E \to \mathbf{Set}$ is pro-representable by a pro-object with small indexing if and only if it is left exact and accessible.

1:13:11 If E has arbitrary limits, a pro-representable functor $E \to \mathbf{Set}$ is representable if and only if it commutes with these limits.

1:14:14 Between cocomplete accessible categories, F has a right adjoint if and only if it commutes with colimits, and a left adjoint if and only if it commutes with limits and is accessible.

1:15:31 Example (SGA 4): on **Grp**, a functor with small values, pro-represented by a non-small system of simple groups, commutes with limits without being accessible.

37 Morphisms of algebraic theories

1:17:44 A morphism of algebraic theories $T \to T'$ is equivalent to a functor $S' \to S$ commuting with colimits; its right adjoint $S \to S'$ is $T(\mathbf{Set}) \to T'(\mathbf{Set})$.

1:20:53 The functors $S \to S'$ arising from morphisms of algebraic theories are exactly those that commute with limits and are accessible.

1:22:11 If $F \to I$ is fibred over I small, with accessible fibres and transition functors, the category of cartesian sections (2-limit) and that of all sections are accessible.

1:24:21 Corollary: for E accessible and I small, $\mathrm{Hom}(I, E)$ is accessible, as is $\mathrm{Hom}(I[I^{-1}], E)$, the functors sending every arrow to an isomorphism.

1:28:40 The 2-colimit of a small system of cocomplete accessible categories, taken in cocomplete categories and cocontinuous functors, is accessible.

1:31:03 Question: is the 2-colimit in **Cat** of a small system of accessible categories accessible?

1:32:09 Example: the coproduct in **Cat** of two nonempty categories with colimits has no initial object, nor a coproduct of an object of each.

1:45:06 For a morphism of algebraic theories, $T(\mathbf{Set}) \to T'(\mathbf{Set})$ has a left adjoint (free structure); example: the enveloping group of a monoid.

1:47:14 If a fully faithful functor admits a left adjoint, this adjoint is a localization (passage to a category of fractions).

1:48:19 The full subtheories of $T_{R'}$ correspond to the localizations $R' \to R'[\Sigma'^{-1}]$ in Δ, Σ' stable under limits, and to the reflective full subcategories of S' stable under limits.

1:50:27 For an algebraic subtheory, Σ' is generated, by limit operations, by a small set of arrows.

1:51:31 Question: is a full subcategory S of an accessible category S', stable under limits and whose inclusion admits a left adjoint, accessible?

38 Constructing algebraic theories

Tape 3, side A

1:57:30 Limit-passage clause: the 2-limit, over T^I, of a filtered system of algebraic theories with base I is algebraic with base I.

1:58:34 Representable theories form a full 2-subcategory of the Δ-theories, 2-equivalent to Δ^{op} via $R \mapsto T_R$; it contains the algebraic theories.

2:00:40 R is algebraic with (discrete) base I if it is the end of a transfinite sequence in Δ starting from the envelope of I, each step inverting, equalizing or adding arrows, with 2-colimits at limit ordinals.

2:03:54 Faithfulness of $\mathrm{Hom}_\Delta(R, C) \to C^I$ for every C does not suffice for $I \to R$ to be basic (known counterexample).

2:03:54 "Problem three" (open): is $I \to R$ basic if and only if every strictly full subcategory of R stable under limits of type D and containing the image of I is equal to R?

2:06:04 This condition is equivalent to: the transfinite saturation of the image of I under limits of type D' and D'' reaches R.

2:08:14 If D' and D'' are essentially small (and if problems one and two preserve smallness), the algebraic objects of Δ are exactly the small ones.

2:10:23 Conjectural statement: if R is algebraic, one can enlarge I so that no arrows remain to be added; T_R is then a full subtheory of $C \mapsto \mathrm{Hom}(I', C)$.

39 Representability of theories

2:17:56 Necessary condition for 2-representability: T commutes with 2-limits, $T(\varprojlim C_i) \simeq \varprojlim T(C_i)$.

2:18:59 On a category with arbitrary limits, a functor to **Set** that commutes with these limits is pro-representable by a projective system, possibly non-small.

2:20:10 If T is 2-representable, T sends faithful (resp. fully faithful) Δ-functors to faithful (resp. fully faithful) functors.

40 Exactness properties of theories

2:23:27 Exactness property given by $R \in \Delta_D$ and Σ: every Δ_D-functor $R \to C$ inverts Σ; equivalent to: $\mathrm{Hom}_{\Delta_D}(R', C) \to \mathrm{Hom}_{\Delta_D}(R, C)$ is an equivalence, R' the localization.

2:25:38 "Generalized exactness properties": for a given Δ_D-functor $R \to R'$, require that $\mathrm{Hom}(R', C) \to \mathrm{Hom}(R, C)$ be an equivalence, fully faithful or faithful.

2:26:42 The 2-subcategories of Δ_D so defined are stable under 2-limits in **Cat**.

2:28:51 Example ($D = \varnothing$): requiring the arrow of $\{0 \to 1\}$ to be inverted defines groupoids.

2:29:58 Equalizing the two arrows of $\{0 \rightrightarrows 1\}$ defines preordered categories, equivalent to preordered sets, 2-equivalent to ordered sets.

2:32:06 The two conditions together define essentially discrete categories, 2-equivalent to **Set**.

2:33:10 For $\Delta =$ **Cat**: the envelope of I is I; problem two is solved by the category of fractions $I[\Sigma^{-1}]$ (Gabriel–Zisman).

2:34:16 For groupoids: the envelope of I is its fundamental groupoid $I[I^{-1}]$, with the same connected components; problem two is trivial.

2:36:21 For preorders: the envelope of I is $\mathrm{Ob}\, I$ with the preorder $x \geq y$ if there is an arrow $x \to y$.

2:37:24 Making relations $x_i \leq y_i$ of a preorder invertible: take the preorder generated by the given preorder and the $y_i \leq x_i$.

2:38:30 For discrete categories: the envelope of I is $\pi_0(I)$, the set of connected components; problem two is trivial.

2:41:54 If $T = T_R$ and if C has limits of type δ commuting with colimits of type D'', $T(C)$ has these limits, computed argumentwise.

Tape 3, side B

2:45:15 A morphism of representable theories $T \to T'$ gives $T(C) \to T'(C)$ commuting with these limits; likewise $T(C) \to T(C')$ if $C \to C'$ preserves them.

2:46:18 For every object r of R, the evaluation functor $r_C : T(C) \to C$ commutes with limits of type δ.

2:47:20 Example: the category of torsors under a group $G \neq 1$ has no products, no coproducts, and no initial or terminal object.

2:48:23 If $D'' = \varnothing$ and if C has limits of type δ (resp. arbitrary), $T(C)$ has them too.

2:50:40 For $\Delta' \subset \Delta$ (fewer limits required), restriction sends Δ-theories to Δ'-theories and algebraic theories to algebraic theories.

2:52:50 If problems one and two have a solution in Δ', the restriction of T_R to Δ' is represented by the "Δ–Δ' envelope" R' of R.

2:56:04 R' is obtained from the envelope of R in Δ' by inverting the comparison arrows of the limits of type D of R.

3:01:55 The restriction functor $\mathrm{Hom}(T_1, T_2) \to \mathrm{Hom}(T_1', T_2')$ is in general neither faithful, nor full, nor essentially surjective.

41 Envelopes and categories of fractions

3:10:48 For J small, $J \to \widehat{J}$ is the envelope under colimits: for D cocomplete, restriction of cocontinuous functors $\widehat{J} \to D$ to J is an equivalence.

3:11:49 Every presheaf F on J is the colimit over J/F of the objects of J over F; the extension is $\widehat{\varphi}(F) = \varinjlim_{J/F} \varphi(X)$.

3:14:59 Problem one for $\Delta =$ categories with small limits: the envelope of a small I is $\mathrm{Hom}(I, \mathbf{Set})^{\mathrm{op}}$, with the Yoneda embedding.

3:17:04 In Δ, equalizing $f, g : X \rightrightarrows Y$ amounts to inverting the kernel $\mathrm{Ker}(f, g) \to X$.

3:18:06 Problem two: if Σ' is the saturation of Σ, the category of fractions $C[\Sigma'^{-1}]$ taken in **Cat** is in Δ, and $C \to C[\Sigma'^{-1}]$ is the solution.

3:21:33 The saturation Σ' (arrows inverted by every Δ-functor inverting Σ) contains the isomorphisms, is stable under composition and base change, and if two of u, v, vu are in it, so is the third.

3:23:39 If $u : X \to Y$ is in Σ', the diagonal $X \to X \times_Y X$ is in Σ'.

3:24:44 Σ' admits a calculus of fractions: $\mathrm{Hom}(X, Y) = \varinjlim \mathrm{Hom}_C(X', Y)$, over the filtered projective system of the $X' \to X$ in Σ'.

3:27:53 If C has finite limits, so does $C[\Sigma'^{-1}]$, and the localization functor is left exact.

3:28:59 Σ' is stable under arbitrary fibre products: if the $X_i \to Y$ are in Σ', so is $\prod_Y X_i \to Y$.

3:31:07 $C[\Sigma'^{-1}]$ has arbitrary products, images of those of C; thus $C \to C[\Sigma'^{-1}]$ is a Δ-functor.

42 Further remarks on algebraic theories

Session of 10 July 1973

Video 3/4 (4gynb9uTEE4): tape 1, side A (3:31:57–4:36:29); tape 1, side B (4:36:29–5:41:33).

Tape 1, side A

3:35:28 For E in Δ and a non-small saturated class $\widetilde{\Sigma}$, the category of fractions $E[\widetilde{\Sigma}^{-1}]$ may have non-small Homs, and is then not in Δ.

3:37:36 For an algebraic theory R with base objects I, I is not in general strictly cogenerating in R: a group structure is not determined by the maps $x \mapsto x^n$.

3:39:45 Strict generation is not transitive: $\mathbb{Z}$ is strictly generating in free groups, these are so in **Gr**, but $\{\mathbb{Z}\}$ is not in **Gr** (likewise in **Ab**).

3:41:54 An algebraic theory relative to limits is not always obtained in two steps (arrows, then axioms); a transfinite construction, indexed by a small ordinal, may be needed.

3:43:00 If $R \in \Delta$ is algebraic, R admits a small strictly cogenerating family; the converse is false.

3:44:08 Example: $\mathbf{Set}^{\mathrm{op}}$ admits a strict generator (the two-element set) but is the category $T(\mathbf{Set})$ of no algebraic theory T.

3:46:17 $\mathbf{Set}^{\mathrm{op}}$ is equivalent to the category of complete atomic Boolean algebras, via $X \mapsto \mathcal{P}(X)$.

3:52:47 Completeness of a Boolean algebra concerns families indexed by an arbitrary set I: a non-small family of operations and axioms, excluded from the algebraic theories of the course.

4:07:05 Every complete Boolean algebra falls, for a suitable cardinal π, under the theory of π-complete Boolean algebras, which has only a small set of operations.

4:10:16 The objects of Δ representing algebraic theories are the Δ-envelopes (adjoining small limits) of small categories of Δ_π (limits of cardinality $\leq \pi$).

4:11:21 The categories $S = R^{\mathrm{op}} = T(\mathbf{Set})$ of set-theoretic realizations of these theories are accessible categories.

43 Accessible categories

4:14:44 Definition: a preordered set I "dominates π" if it is filtered and every family of cardinality $\leq \pi$ of elements of I has an upper bound; for $\pi \geq 2$, the second condition implies the first.

4:15:52 Definition: E satisfies L_π if colimits indexed by every small I dominating π exist in E; E satisfies L if it satisfies L_π for some small cardinal π.

4:18:02 Definition: $F\colon E \to E'$ is π-accessible if E satisfies L_π and F commutes with colimits indexed by the I dominating π; accessible if it is π-accessible for some small π.

4:21:13 Definition: an object X of E is π-accessible (resp. accessible) if the functor $\mathrm{Hom}(X, -)$ from E to **Set** is π-accessible (resp. accessible).

4:23:29 Definition: E is π-accessible if it has a small full strictly generating subcategory C consisting of π-accessible objects (every object is a colimit of a diagram in C).

4:24:37 This definition is stronger than that of SGA 4, I, 9, which only required a small generating subcategory.

4:26:48 If E, F satisfy L_π, the full subcategory $\mathrm{Hom}_\pi(E, F)$ of π-accessible functors is stable in $\mathrm{Hom}(E, F)$ under colimits computed argumentwise.

4:27:56 It is also stable under limits, computed argumentwise, indexed by a small category J with at most π arrows.

4:30:13 In **Set**, colimits along an I dominating π commute with limits along a category with at most π arrows.

4:30:13 Accessible functors form a full subcategory of $\mathrm{Hom}(E, F)$ stable under small colimits and limits computed argumentwise.

4:31:16 If E satisfies L_π, the full subcategory E_π of π-accessible objects is stable under colimits indexed by a category with at most π arrows.

4:32:21 If E satisfies L, the full subcategory of accessible objects of E is stable under small colimits.

4:33:26 If E is accessible, every object of E is accessible.

44 Examples of accessible categories

Tape 1, side B

4:36:32 A small subcategory C is generating by universal strict epimorphisms if every object is the target of a universally strictly epimorphic family from C; this is stronger than strictly generating.

4:37:34 In a topos, every epimorphic family is strictly epimorphic and universal.

4:37:34 If E satisfies L, has fibre products, kernels $(u, v) \mapsto \mathrm{Ker}(u, v)$ forming an accessible functor, and such a C, then every object of E is accessible and E is accessible (SGA 4, I, 9.11).

4:39:42 Example: every topos is an accessible category.

4:39:42 A set X of cardinality ≥ 2 is not an accessible object of $\mathbf{Set}^{\mathrm{op}}$; hence $\mathbf{Set}^{\mathrm{op}}$, which has small colimits and a generator, is not accessible.

4:41:48 For π infinite, the subsets of cardinality $\leq \pi$ of a set I form an ordered set dominating π, and $\prod_{i\in I} Y_i$ is the limit of the corresponding partial products.

4:45:02 If card $I > \pi$ and card $Y_i \geq 2$, the indicator of a point of $\prod Y_i$ factors through no partial product; the opposite of a nondegenerate topos, or of **Ab**, is not accessible.

4:48:17 Question: if E and E^{op} are accessible, is E equivalent to the point category?

45 Ind-categories

4:50:29 Definition: for C small, $\mathrm{Ind}(C)$ is the category of filtered inductive systems in C, identified via $\varinjlim$ with a full subcategory of $\widehat{C}$.

4:52:37 $\mathrm{Hom}_{\mathrm{Ind}(C)}((X_i),(Y_j)) = \varprojlim_i \varinjlim_j \mathrm{Hom}(X_i, Y_j)$ (SGA 4, I, 8).

4:54:44 $C \to \mathrm{Ind}(C)$ is universal: for E with filtered colimits, restriction to C of the functors $\mathrm{Ind}(C) \to E$ commuting with these colimits is an equivalence with $\mathrm{Hom}(C, E)$.

4:58:00 If C has finite colimits, $\mathrm{Ind}(C)$ has small colimits and, for E having them, $F \mapsto F|_C$ identifies the functors commuting with these colimits with the right exact functors $C \to E$.

5:00:08 Definition: $\mathrm{Ind}_\pi(C)$ is the full subcategory of $\mathrm{Ind}(C)$ of the systems indexed by an I dominating π; for C small, it is π-accessible.

5:05:31 An object of $\mathrm{Ind}_\pi(C)$ is π-accessible if and only if it is a direct summand of an object of C; a direct summand of a π-accessible object is again so.

5:06:36 Definition: C is Karoubian if every idempotent has an image summand; $\mathrm{Ind}(C)$ and $\mathrm{Ind}_\pi(C)$ do not change if C is replaced by its Karoubian envelope.

5:07:40 If C is Karoubian and $E = \mathrm{Ind}_\pi(C)$, then $C \simeq E_\pi$ and $E \simeq \mathrm{Ind}_\pi(E_\pi)$.

46 Cardinal filtrations

5:08:44 For $\pi' \geq \pi$, $\mathrm{Fil}^{\pi'} = \mathrm{Ind}_{\pi,\pi'}(C)$ (systems indexed by an I dominating π with card $I \leq \pi'$) filters $\mathrm{Ind}_\pi(C)$ by essentially small subcategories, and $\mathrm{Fil}^\pi \simeq C$.

5:09:49 The objects of $\mathrm{Fil}^{\pi'}$ are π'-accessible; if $\pi'^\pi = \pi'$, every π'-accessible object of $\mathrm{Ind}_\pi(C)$ is a direct summand of an object of $\mathrm{Fil}^{\pi'}$.

5:10:52 "Cardinal filtration": a family $(\mathrm{Fil}^{\pi'})_{\pi'\geq\pi}$ of essentially small subcategories, stable under $\varinjlim$ of index I dominating π, card $I \leq \pi'$, every object of $\mathrm{Fil}^{\pi''}$ being a colimit of objects of $\mathrm{Fil}^{\pi'}$ with index dominating π', of cardinality $\leq \pi''$.

5:14:07 The $\mathrm{Ind}_{\pi,\pi'}(C)$ form a cardinal filtration of $\mathrm{Ind}_\pi(C)$ if one restricts to cardinals such that $\pi''^{\pi'} = \pi''$ (bound $\pi''^{\pi'}$ of SGA 4, I, 9).

5:15:11 Theorem: if E has small colimits, the following are equivalent: a) E is π-accessible; b) $E \simeq \mathrm{Ind}_\pi(C)$, C small; c) E_π is essentially small and strictly generating.

5:22:51 If $C \subset E_\pi$ is small and full, the extension $\mathrm{Ind}_\pi(C) \to E$ is fully faithful; it is an equivalence if C is strictly generating and stable under limits with at most π arrows.

5:23:53 A colimit indexed by a small category I is the limit, along the ordered set (dominating π) of subcategories of I with at most π arrows, of the partial limits.

5:28:22 If E is π-accessible and has small colimits, $E \simeq \mathrm{Ind}_{\pi'}(E_{\pi'})$ for every $\pi' \geq \pi$.

5:31:52 A π-accessible functor does not necessarily send π-accessible objects to π-accessible objects (example: E a point).

5:32:58 If E, E' are π_0-accessible with small colimits, $F\colon E \to E'$ is accessible if and only if there exist $\pi \geq \pi_0$ and $f\colon E_\pi \to E'_\pi$ with $F \simeq \mathrm{Ind}_\pi(f)$.

47 Strictly generating functors

Session of 11 July 1973

Video 3/4 (4gynb9uTEE4): tape 1 (5:41:33–6:46:44); tape 2, side A (6:46:44–7:37:01).

Tape 1

5:43:46 For $i\colon C \to E$, $\varinjlim_{(Y,u)\in C/X} i(Y) \to X$ is an isomorphism for every X if and only if $E \to \widehat{C}$, $X \mapsto \mathrm{Hom}(i(-), X)$, is fully faithful: i is then called strictly generating.

5:46:59 Then every X is the target of a strictly epimorphic family $i(Y_\alpha) \to X$; the converse holds if i is fully faithful and E has fibre products and universal strictly epimorphic families.

5:50:08 If E has small colimits and C is small, i extends to $i_!\colon \widehat{C} \to E$, left adjoint to $E \to \widehat{C}$; i is strictly generating if and only if $i_!$ makes E a category of fractions of $\widehat{C}$.

5:52:19 A category with small colimits has a small strictly generating subcategory if and only if it is equivalent to a reflective full subcategory of some $\widehat{C}$, C small.

5:53:21 Examples: $\mathbf{Set}^{\mathrm{op}}$ (complete atomic Boolean algebras) and $\mathbf{Ab}^{\mathrm{op}}$ (compact abelian groups) have small colimits and a small strictly generating family.

5:57:44 In the category of compact spaces, the one-point space is not strictly generating: the underlying-set functor is faithful but not fully faithful.

48 Theories defined by finite limits

6:05:23 For Δ_D (limits of type D' only), the envelope of C is the subcategory of $\mathrm{Hom}(C, \mathbf{Set})^{\mathrm{op}}$ generated by C under iterated limits of type D'; C embeds fully in it.

6:07:34 If R_0 represents a theory for Δ_D, the theory extended to Δ is represented by $R = T(\mathbf{Set})^{\mathrm{op}}$, the envelope of R_0 under small limits; $R_0 \subset R$ is full.

6:08:38 A theory relative to Δ_D is equivalent to the pair (S, Σ_0), $S = T(\mathbf{Set})$, $\Sigma_0 = R_0^{\mathrm{op}} \subset S$, where S is the envelope of Σ_0 under small colimits.

6:10:44 For D' consisting of the finite diagrams, the algebraic theories are the small categories R_0 with finite limits; dually, Σ_0 is small with finite colimits.

49 Finitely presented objects

6:12:54 Definition: X is finitely presented ($\aleph_0$-accessible) if $\mathrm{Hom}(X, -)$ commutes with filtered colimits; we write S_{fp} for the full subcategory of these objects.

6:13:58 For a theory defined by finite limits, $\Sigma_0 \simeq S_{\text{fp}}$ and $S \simeq \text{Ind}(\Sigma_0)$; S and Σ_0 determine each other, and $R_0 \simeq (S_{\text{fp}})^{\text{op}}$.

6:15:01 For $\Sigma_0 \subset S$ full, $\text{Ind}(\Sigma_0) \to S$ is fully faithful if and only if $\Sigma_0 \subset S_{\text{fp}}$; if Σ_0 is stable under finite colimits, it is an equivalence if and only if Σ_0 is strictly generating.

6:16:10 Example: in categories of groups or modules, the finitely presented objects are the groups or modules finitely presented in the usual sense.

6:17:18 S with small colimits is $T(\mathbf{Set})$ for a theory defined by finite limits if and only if S_{fp} is essentially small and strictly generating.

6:19:26 In **Gr**, for C full on the object $\mathbb{Z}$, $\widehat{C}$ consists of the sets acted on by the monoid $(\mathbb{Z}, \times)$; $G \mapsto (G, x \mapsto x^n)$ is faithful but not full: $\mathbb{Z}$ is generating, but not strictly.

6:22:40 For $C \subset E$ full, $E \to \widehat{C}$ fully faithful, faithful, conservative, or conservative on monomorphisms: distinct notions of generation, which coincide in a topos.

6:25:47 Finitely generated free groups form a small strictly generating subcategory of **Gr**.

6:26:51 The morphisms $T \to T'$ of theories defined by finite limits correspond to the functors $S' \to S$ commuting with small colimits and preserving finitely presented objects.

50 Lawvere theories and finite products

6:31:05 For finite products, the envelope of an object X is $(\mathbf{Set}_{\text{f}})^{\text{op}}$, with objects X^n; for products of cardinality $\leq \pi$, the opposite of the sets of cardinality $\leq \pi$.

6:34:22 A theory with one base object defined by finite products is a category with objects $\mathbb{N}$, n being the n-th power of 1, given by the $H_n = \text{Hom}(X^n, X)$ and their substitution (Lazard's "analyseurs").

6:36:31 The realizations of such a (Lawvere) theory in a category E with finite products are the functors to E commuting with finite products.

6:37:33 For a set I of base objects, one starts from the opposite of the category of finite sets over I (objects $\prod_i X_i^{n_i}$, n_i almost all zero).

6:38:37 For a theory defined by finite products, Σ_0 is stable under finite coproducts and strictly generating in S, but this does not characterize the pairs (S, Σ_0).

6:40:44 S with small limits is the category of models of a theory with finite products and base objects I if and only if its equivalence relations are effective and I, consisting of finitely presented objects, detects effective epimorphisms ($\text{Hom}(i, f)$ surjective).

6:43:53 Then $S \to \mathbf{Set}^I$, $X \mapsto (\text{Hom}(i, X))_{i \in I}$, has a left adjoint and is monadic ("tripleable").

51 Birkhoff's theorem

Tape 2, side A

6:50:08 Birkhoff's theorem: a full subcategory of a category of structures defined by finite products, stable under products, subobjects and quotients, is defined by equations.

6:51:11 Such a subcategory is reflective: the adjoint associates to X its largest quotient belonging to the subcategory.

6:52:17 Counterexample: torsion-free abelian groups, stable under products and subgroups but not under quotients, are not definable by equations in **Ab**.

6:54:26 To a Birkhoff subcategory $S' \subset S$ corresponds $R \to R'$ bijective on objects and surjective on arrows: T' is deduced from T by equalizing arrows.

7:05:11 Groups are the monoids in which a certain arrow is invertible; not stable under submonoids, they are not definable by equations in monoids.

7:07:29 The reflective full subcategories $S' \subset S$ such that the composite $S' \to \mathbf{Set}^I$ is monadic form a larger class than that of Birkhoff subcategories.

52 Pseudo-topoi and classifying topoi

7:10:46 Definition: a "pseudo-topos" satisfies Giraud's axioms a), b), c): small colimits and finite limits; universal disjoint coproducts; universal effective equivalence relations.

7:12:55 The functors considered commute with small colimits and finite limits: they are the inverse images f^* of morphisms of topoi in the opposite direction.

7:14:08 A topos $\mathcal{E}$ classifies a theory T, with universal object $U \in T(\mathcal{E})$, if $f \mapsto f^*U$ is an equivalence $\mathrm{Top}(\mathcal{F}, \mathcal{E})^{\mathrm{op}} \simeq T(\mathcal{F})$ for every $\mathcal{F}$ (2-arrows oriented by the direct images).

7:19:36 For C small with finite limits, the envelope of C among pseudo-topoi is $\widehat{C}$, with the Yoneda embedding.

53 Continuous functors and morphisms of sites

7:21:48 Definition: $f : C \to C'$ between sites is continuous if $F \mapsto F \circ f$ sends sheaves on C' to sheaves on C; this depends only on $\varepsilon_{C'} \circ f : C \to \widetilde{C'}$.

7:22:55 By composition with ε_C, the functors $\widetilde{C} \to \widetilde{C'}$ commuting with small colimits are identified with the continuous functors $C \to \widetilde{C'}$.

7:25:04 If C, C' have finite limits and f commutes with them, f is continuous if and only if it sends covering families to covering families.

7:25:04 Example: for $X' \to X$ continuous, the inverse image of open sets is such a continuous functor from the site of open sets of X to that of X'.

7:26:07 In general, f is continuous if and only if, for every covering sieve R of X, $\varepsilon_{C'} f(X) = \varinjlim_{Y \in C/R} \varepsilon_{C'} f(Y)$.

7:27:12 A sieve R of X is covering if and only if $\varepsilon(X) = \varinjlim_{Y \in C/R} \varepsilon(Y)$ in $\widetilde{C}$.

7:30:21 If C, C' have finite limits and the topology of C' is coarser than the canonical one, a continuous f induces a morphism of topoi if and only if f is left exact.

7:31:25 Definition: a morphism of sites is a continuous functor that induces a morphism of topoi; it is oriented in the direction of that morphism.

7:32:31 For a site C with finite limits and a topos $\mathcal{E}$, $\mathrm{Top}(\mathcal{E}, \widetilde{C})^{\mathrm{op}}$ is equivalent to the left exact functors $C \to \mathcal{E}$ sending covering families to epimorphic families.

7:34:39 $\widehat{C}$ is $\widetilde{C}$ for the coarsest topology (covering families: those having a member with a section).

7:34:39 If C has finite limits, $\mathrm{Top}(\mathcal{E}, \widehat{C})^{\mathrm{op}}$ is equivalent to the left exact functors $C \to \mathcal{E}$.

54 Classifying topoi of finite-limit theories

Session of 12 July 1973

Video 4/4 (0RG14LNb5gE): tape 1 (0:00:00–0:49:57); tape 2, side A (0:49:57–1:38:22); tape 2, side B (1:38:22–2:26:53); tape 3, side A (2:26:53–3:15:55); tape 3, side B (3:15:55–4:05:06); tape 4, side A (4:05:06–4:53:49); tape 4, side B (4:53:49–5:36:07).

Tape 1

0:00:05 If C is a small category with finite limits, $\widehat{C}$ classifies the theory T_C: $\mathrm{Hom}_{\mathrm{Top}}(\mathcal{E}, \widehat{C}) \simeq \mathrm{Hom}_{\mathrm{lex}}(C, \mathcal{E})$ for every topos $\mathcal{E}$.

0:02:11 Example: the classifying topos of groups is the category of covariant functors from finitely presented groups to **Set**.

0:03:15 Example: the classifying topos of commutative rings is that of functors from finitely presented rings to **Set**, i.e. of presheaves on affine schemes of finite type over $\mathbb{Z}$.

0:05:25 The envelope $\widetilde{C}$ of C under finite limits is the opposite of the category of finitely presented functors $C \to$ **Set**; $C \to \widetilde{C}$ is fully faithful.

0:06:28 If $\mathcal{E}$ has finite limits, $\mathrm{Hom}(C, \mathcal{E}) \simeq \mathrm{Hom}_{\mathrm{lex}}(\widetilde{C}, \mathcal{E})$.

0:07:34 Enveloping topos of a small category C: the topos of presheaves on $\widetilde{C}$; the morphisms from a topos $\mathcal{E}$ to it are equivalent to $\mathrm{Hom}(C, \mathcal{E})$.

55 Subtopoi and localization

0:10:03 Localization: for a set Σ of arrows of a topos $\mathcal{E}$, one seeks a universal Δ-functor $\mathcal{E} \to \mathcal{E}[\Sigma^{-1}]$ making the arrows of Σ invertible.

0:12:16 The subtopoi of $\mathcal{E}$ correspond bijectively to the saturated sets of arrows of $\mathcal{E}$ (stable under limits and local in nature) (SGA 4, exercise).

0:13:21 Definition: a morphism of topoi $i\colon \mathcal{E}' \to \mathcal{E}$ is an embedding if i_* is fully faithful, i.e. if i^* is a localization functor.

0:15:30 The subtopoi of $\mathcal{E}$ are identified with the strictly full subcategories whose inclusion admits a left exact left adjoint.

0:16:37 For a small category C, the topologies J on C correspond bijectively, via $J \mapsto \mathrm{Sh}(C, J)$, to the subtopoi of $\widehat{C}$.

0:17:39 A continuous map $f\colon X \to Y$ induces an embedding of topoi iff the topology of X is the inverse image of that of Y; subspaces of Y give subtopoi, but not all of them.

0:18:44 A morphism $\mathcal{F} \to \mathcal{E}$ factors through a subtopos $\mathcal{E}'$ iff its direct image factors through $\mathcal{E}'$, iff its inverse image factors through the localization $\mathcal{E} \to \mathcal{E}'$.

0:19:49 To a subtopos $\mathcal{E}'$ one associates the arrows made invertible by i^*; to Σ, the objects X such that $\mathrm{Hom}(Z, X) \to \mathrm{Hom}(Y, X)$ is bijective for every $Y \to Z$ in Σ.

0:22:00 $\mathcal{E}[\Sigma^{-1}]$ exists and is a subtopos of $\mathcal{E}$; every subtopos is obtained in this way by inverting a family of arrows.

0:23:02 If $\mathcal{E} = \mathrm{Sh}(C, J)$, its subtopoi correspond to the topologies on C finer than J; likewise for the subtheories of the theory represented by $\mathcal{E}$.

56 Generating subcategories of a topos

0:32:47 If a small full subcategory C generates the topos $\mathcal{E}$, $\mathcal{E}$ is a subtopos of $\widehat{C}$: $\mathcal{E} \simeq \mathrm{Sh}(C, J)$ for the induced topology.

0:37:28 For C full in $\mathcal{E}$: (a) $\mathcal{E} \to \widehat{C}$ fully faithful is equivalent to (b) every X is a colimit of the Y in C/X; (b) implies (c): every X is the target of a strictly epimorphic family from C.

0:39:40 Counterexample to (c) $\Rightarrow$ (b): in groups, with $C = \{\mathbb{Z}\}$, $\mathbb{Z} * \mathbb{Z}$ satisfies (c) but is not a colimit of $C/(\mathbb{Z} * \mathbb{Z})$.

0:40:46 Enlarging a strictly epimorphic family can make it lose this property (all the arrows $\mathbb{Z} \to \mathbb{Z} * \mathbb{Z}$ in groups).

0:42:52 If fibre products exist and distribute over coproducts, $(X_i \to X)$ is strictly epimorphic iff $\coprod X_i \to X$ is; this is false in general.

0:46:05 Corrected statement: (c′) every X is the target of a family from C that is strictly and universally epimorphic; then (c′) implies (b).

0:47:09 If a strictly and universally epimorphic family factors through a family $(X_i \to Y)$, the latter is again strictly and universally epimorphic.

0:48:11 (b) is equivalent to (b′): for every X, the family of all arrows $Y \to X$, Y in C, is strictly epimorphic.

Tape 2, side A

0:50:01 A family $(Y \to X)$ is strictly epimorphic iff for every Z, $\mathrm{Hom}(X, Z)$ is identified with the compatible families of elements of the $\mathrm{Hom}(Y, Z)$ (sheaf condition).

57 Algebraic theories and topoi

1:04:03 Theorem: $R \in \Delta$ represents an algebraic theory iff R is a topos; $B\colon I \to R$ is basic iff the finite products of the $B(i)$ generate R.

1:06:58 Every topos R represents an algebraic theory: if $C \subset R$ is small, generating and stable under finite limits, T_R is a full subtheory of $T_C = \mathrm{Hom}_{\mathrm{lex}}(C, -)$.

1:10:36 For such a C, the morphisms of topoi $\mathcal{E} \to R$ correspond to the left exact functors $C \to \mathcal{E}$ sending covering families to epimorphic families.

1:11:47 If $R' \to R$ induces for every topos $\mathcal{E}$ a fully faithful embedding $T_{R'}(\mathcal{E}) \to T_R(\mathcal{E})$, R' is a subtopos of R (exercise of SGA 4).

1:19:23 A subtopos of $R = \mathrm{Sh}(C, \tau)$ is $\mathrm{Sh}(C, \tau')$ with τ' finer; its theory is deduced from T_R by a small set of axioms.

58 Limits of topoi

1:23:42 Arbitrary (not necessarily filtered) 2-limits of topoi exist; in Δ, they are 2-colimits.

1:29:00 Construction: $C_i \subset R_i$ compatible small generating subcategories, C the 2-colimit of the C_i in **Cat**, then localization of the enveloping topos of C at a small Σ.

1:30:05 The 2-colimit of a system of small categories is obtained by inverting the cartesian arrows of the associated (co)fibred category.

1:36:31 A 2-limit of accessible categories, with accessible transition functors, is accessible.

59 Algebraic theories built from finite products

Tape 2, side B

1:38:26 The category J_0 freely generated by finite products over a set I is the opposite of the families of finite sets indexed by I, almost all empty; it is also the envelope under finite limits of the discrete I.

1:40:37 $R_0 = \widehat{J_0}$ classifies families of objects indexed by I: $\mathrm{Hom}_\Delta(R_0, \mathcal{E}) \simeq \mathcal{E}^I$.

1:41:40 Adding axioms: $R_{\alpha+1} = R_\alpha[\Sigma_\alpha^{-1}]$ has the same objects as R_α, so J_0 still generates it.

1:43:52 Adding arrows given by a graph $D = (H \rightrightarrows S)$: $R_{\alpha+1}$ is the 2-pushout in Δ of R_α and $\rho(\widetilde{D})$ over $\rho(S)$ (ρ the enveloping topos, $\widetilde{D}$ the free category).

1:51:23 In this construction, every object of $R_{\alpha+1}$ is covered by images of objects of R_α; hence J_0 generates $R_{\alpha+1}$.

1:56:57 Limit step: for α a limit, R_α is the 2-colimit in Δ of the R_β, $\beta < \alpha$; J_0 still generates it.

1:59:07 Necessity: if $I \to R$ is basic for an algebraic theory, R is a topos generated by the finite products of objects of I.

2:00:13 Sufficiency: if the finite products of the objects of I generate the topos R, the theory is obtained without transfinite recursion: finite products, arrows between them, then axioms.

2:03:18 For J small with finite products, the theory $\mathcal{E} \mapsto \mathrm{Hom}_\Pi(J, \mathcal{E})$ of functors preserving finite products is represented by a topos $\rho_{\Pi,\Delta}(J)$.

2:08:52 If the subcategory $J_2 \subset R$ of finite products of objects of I generates R, R is a subtopos of $\rho_{\Pi,\Delta}(J_2)$; two successive localizations make it one.

2:16:31 Preservation of finite products is an axiom: the arrow of constructions $\varepsilon_{X\times Y} \to \varepsilon_X \times \varepsilon_Y$ must become an isomorphism.

2:21:54 If $R \in \Delta$ represents a theory, its category of constructions is R; equalizing two arrows amounts to inverting a kernel, hence a single type of axiom.

Tape 3, side A

2:28:03 In a topos, $u\colon X \to Y$ is an epimorphism iff $Y \sqcup_X Y \to Y$ is an isomorphism; "being covering" is thus written as an invertibility axiom.

60 Classifying topos of an object and its subtopoi

2:33:54 The classifying topos of an object without structure is the category of functors from finite sets to **Set** (presheaves on $J = \{X^n\}$).

2:36:05 If B_T classifies the theory T, its category of points is equivalent to $T(\mathbf{Set})$.

2:37:13 For all topoi $\mathcal{E}$ and B, $\mathrm{Hom}_\Delta(B, \mathcal{E})$ admits filtered colimits, since in a topos they commute with finite limits.

2:39:28 The essential points of the classifying topos of objects correspond to finite sets.

2:40:31 The smallest subtopos containing a family of points of a topos is defined by the arrows whose stalks at these points are isomorphisms.

2:42:38 The subobjects of the terminal object of the classifying topos B of objects are $\varnothing$, the image of $X \to e$, and e: B has three open subtopoi.

2:44:48 The closed complement Φ_0 of B^* classifies initial objects; it is equivalent to the point topos, whose point is the empty set.

2:46:57 The open B^* classifies the objects x such that $x \to e$ is an epimorphism; its points are the nonempty sets.

2:49:05 The empty subtopos classifies the theory that has a model only in the empty topos.

2:50:10 The subtopos Φ_{01} classifies the subobjects of the terminal object ($x \to e$ a monomorphism); its points are the sets with at most one element.

2:52:22 $\Phi_1 = \Phi_{01} \cap B^*$ classifies terminal objects; Φ_0 and Φ_1 are equivalent to the point topos but are two distinct points of B.

2:55:37 $\Phi_{01} \simeq \widehat{\Delta_1}$, where $\Delta_1 = \{0 \to 1\}$: the morphisms $\mathcal{E} \to \widehat{\Delta_1}$ correspond to the monomorphisms $x \hookrightarrow e$.

2:57:45 $\widehat{\Delta_1}$ is the topos of the Sierpiński space (spectrum of a discrete valuation ring); a morphism to it is equivalent to an open of the source topos.

2:58:47 Every T_0 topological space embeds in a product of copies of the Sierpiński space.

2:59:53 The subtopoi of a topos form a complete lattice (arbitrary intersections and suprema).

61 Flat functors and torsors

3:06:38 In the opposite of finitely presented groups, the universal group is $\mathbb{Z}$ with its cogroup structure; the classifying topos of groups is its topos of presheaves.

3:10:03 $\widehat{C}$ classifies the functors $C \to \mathcal{E}$ whose continuous extension to $\widehat{C}$ is left exact (flat functors); if C has finite limits, these are the left exact functors.

3:12:13 For a group G viewed as a category C with one object, $\widehat{C} = B_G$; a functor $C \to \mathcal{E}$ is an object X with a G-action, extended by $P \mapsto P \wedge^G X$.

3:13:21 Contracted product: $P \wedge^G X$ is the quotient of $P \times X$ by the diagonal action of G, $(pg, x) \sim (p, gx)$.

3:14:27 This extension is left exact iff X is a torsor under the constant group $G_{\mathcal{E}}$: B_G classifies G-torsors.

62 The classifying topos of a group

Tape 3, side B

3:15:58 Problem: for a group G of a topos S, represent on topoi X over S the 2-functor of torsors under the inverse image of G.

3:22:20 The objects of S equipped with a left action of G form a topos B_G: the forgetful functor $B_G \to S$ is faithful and commutes with all limits.

3:24:28 $p^*\colon S \to B_G$ equips an object with the trivial action; a group morphism $G \to G'$ induces by restriction an essential morphism $B_G \to B_{G'}$.

3:26:37 Universal torsor: G acting on itself by left translations (an object of B_G), with the right action of $\overline{G} = p^*G$ by right translations.

3:33:18 Universality: for $q\colon X \to S$, the q^*G-torsors on X correspond to the morphisms $X \to B_G$ over S, via the contracted product.

63 Classifying topoi: further examples

3:39:47 On a stack, the category of objects locally isomorphic to a section s, with their isomorphisms, is equivalent to that of torsors under $\underline{\mathrm{Aut}}(s)$.

3:40:51 Locally free $\mathcal{O}_X$-modules of rank n, with their isomorphisms, are equivalent to torsors under $\mathrm{GL}_n(\mathcal{O}_X)$.

3:42:59 The topos B_{GL_n} over the classifying topos of rings classifies ringed topoi equipped with a locally free module of rank n (with isomorphisms).

3:44:02 Local rings are classified by the subtopos defined by the Zariski topology on affine schemes of finite type over $\mathbb{Z}$.

3:45:06 Strictly local rings are classified by a smaller subtopos, defined by a finer topology (the étale topology).

3:47:12 Locally constant objects do not come from a representable theory: they are not stable under colimits (totally disconnected spaces).

3:48:16 Covering spaces of rank n of the terminal object do not form an algebraic structure: in **Set**, sets with n elements are not stable under filtered colimits.

3:50:26 With only isomorphisms as morphisms, the objects locally isomorphic to the constant object I_n with n elements are equivalent to torsors under the constant group $\mathfrak{S}_n$.

3:51:31 The topos $B_{\mathfrak{S}_n}$ of $\mathfrak{S}_n$-sets classifies covering spaces of rank n of the terminal object, with their isomorphisms.

64 Connected rings

3:52:37 Connectedness of the spectrum of a ring can be written in any topos (every idempotent is 0 or 1, and $0 \neq 1$) and defines a subtopos of the classifying topos of rings.

3:53:42 "Having exactly n connected components" does not define a theory: the property is not stable under filtered colimits.

3:57:59 A ring locally with n connected components is equivalent to a pair: I locally constant of rank n, and a 0-connected ring A_0 over I, of which A is the direct image.

4:00:09 The functor $(I, A_0) \mapsto A$ is fully faithful if the ring morphisms taken are those inducing isomorphisms on the π_0.

4:03:21 The classifying topos of rings locally with n components is constructed over $B_{\mathfrak{S}_n}$, by classifying a 0-connected ring over the universal covering space I_n.

Tape 4, side A

4:09:38 The classifying topos of a ring over an object I of a topos B exists; if I is a sum of n copies of the terminal object, it is $R^n \times B$, R the classifying topos of rings.

65 Left exact functors and topoi

4:13:58 If $\mathcal{E}$ is a topos and C has finite limits, $C \to \mathcal{E}$ is left exact iff its extension $\widehat{C} \to \mathcal{E}$ commuting with colimits is.

4:15:02 Converse: if $\mathcal{E}$ has a small strictly generating subcategory C stable under finite limits and this property holds, $\mathcal{E}$ is a topos.

4:18:14 The classifying topos of a theory defined by finite limits is connected and acyclic.

4:22:31 Elementary topos: a category with finite limits, cartesian closed, equipped with a subobject classifier.

4:25:44 Faithfulness of a G-action on X is not stable under inverse image: it is not an algebraic structure in the sense of topoi.

66 Topoi without points

4:35:21 A theory represented by a nonempty topos may have no set-theoretic model: there exist topoi without points.

4:36:27 Example (Deligne): Borel sets of $[0, 1]$ modulo null measure, countable coverings up to null measure; the resulting topos has no point.

67 The classifying topos of fields

4:40:53 Classifying topos of fields: affine schemes of finite type over $\mathbb{Z}$, covering families generated by $\{D(f) \to X,\ V(f) \to X\}$ and by $X_{\text{red}} \to X$.

4:41:59 For X quasi-compact and quasi-separated, the constructible topology (opens: unions of constructible subsets) makes X compact, with the same points.

4:43:01 For $X = \operatorname{Spec} A$, the colimit of the rings of functions of the finite constructible partitions (reduced structures) has as spectrum X with the constructible topology, with stalks the residue fields.

4:44:06 The classifying topos of fields is identified with presheaves on these compact schemes (constructible morphisms) turning finite sums into products; the universal field is the ring of functions.

68 Flatness

4:45:11 Absolutely flat rings: rings all of whose modules are flat; these are the reduced rings of Krull dimension zero.

4:47:21 Flatness of a module over a ring of a topos is stable under inverse image by a morphism of topoi (Deligne).

4:48:27 Criterion: M is flat over A if for every U and every finitely generated ideal $J \subset A_U$, $M_U \otimes J \to M_U$ is a monomorphism.

4:49:31 This criterion is a countable family of axioms: for each n, on $U_n = A^n$, the ideal generated by the n universal projections.

Tape 4, side B

4:55:00 For a relative scheme of finite presentation over a ringed topos, two notions of flatness (Hakim): a flat morphism of ringed topoi, or locally arising from a flat morphism of schemes.

69 Hypertopoi

5:00:19 In **Set**, an arbitrary product of epimorphisms is an epimorphism (analogue of total disconnectedness); this does not hold in every topos.

5:01:24 The morphism $\mathcal{E} \to$ pt is essential iff the constant sheaf functor commutes with arbitrary products (analogue of local connectedness).

5:02:29 The topoi $\widehat{C}$ have these two properties; that they are the only ones is stated without proof. "Hypertopos": these topoi, with essential morphisms.

5:06:51 A small Karoubian category D is determined by $\widehat{D}$: it is the category of its essential points.

5:10:04 The category of points of $\widehat{C}$, C with finite limits, is $\mathrm{Hom}_{\mathrm{lex}}(C, \mathbf{Set}) \simeq \mathrm{Ind}(C^{\mathrm{op}})$.

5:11:18 For hypertopoi, the only candidate classifying object of T_C is the topos of presheaves on the category of points of $\widehat{C}$, not small in general ("pseudo-hypertopos").

70 Universes

5:15:36 The Weil conjectures are proved by Deligne (1973), within étale cohomology founded on universes.

5:16:37 The universe axiom is equivalent to: every cardinal is bounded by a strongly inaccessible cardinal; the existence of such a cardinal is not provable in ZFC.

5:17:41 "Galaxy": a set satisfying the universe axioms, except stability under unions of families indexed by one of its elements, which is weakened.

71 The standard conjectures

5:32:49 The standard conjectures are formulated on algebraic cycles modulo numerical equivalence, with no particular cohomology; they include the equality of homological and numerical equivalence.

5:33:55 The standard conjectures would allow one to construct motives from algebraic cycles alone, independently of any cohomology.

References

Colin McLarty, *Grothendieck's 1973 topos lectures*, talk, CultureMath (DMA, École normale supérieure, Paris), video published 7 October 2018, https://www.youtube.com/watch?v=5AR55ZsHmKI.

Readings recommended in the lectures

What Grothendieck tells the audience to read, with the moment he says it (video/timestamp). A "?" marks a title the captions do not make certain.

- R. Godement, *Topologie algébrique et théorie des faisceaux*, Hermann, 1958: "the standard reference" for sheaves (1/4, 0:39:18; again 1:59:40).
- G. E. Bredon, *Sheaf Theory*, McGraw-Hill, 1967: mainly cohomological (1/4, 0:40:24).
- M. Artin, A. Grothendieck, J.-L. Verdier, *SGA 4, Théorie des topos et cohomologie étale des schémas*, LNM 269, 270, 305, Springer, 1972–1973: exp. I on universes, generators and ind-objects, § 9 on accessible functors, which he says is told "somewhat inadequately" there (1/4, 2:59:16, 9:59:51; 3/4, 3:45:13, 4:24:37, 4:53:40); exp. V for cohomology (1/4, 7:16:42); the exercises (1/4, 8:16:56, 9:08:54).
- M. Artin, B. Mazur, *Etale Homotopy*, LNM 100, Springer, 1969: homotopy invariants of a topos (1/4, 7:23:07, 9:06:45).
- J. Giraud, *Cohomologie non abélienne*, Springer, 1971: non-abelian cohomology and stacks (1/4, 7:39:09).
- A. Grothendieck, "Sur quelques points d'algèbre homologique", *Tôhoku Math. J.* 9 (1957): cohomology with operators $H^i(X; G, M)$ (1/4, 9:27:02).
- P. Gabriel, M. Zisman, *Calculus of Fractions and Homotopy Theory*, Springer, 1967: categories of fractions (3/4, 2:34:16).
- F. W. Lawvere, on algebraic theories (? title not heard; presumably *Functorial Semantics of Algebraic Theories*, Columbia, 1963), and "a paper of mine" for tripleability (3/4, 6:46:04).